Jiashu Yu

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EDUCATION BACKGROUND

Xingjian College,Tsinghua University ***Sep. 2022–Jun. 2027 (*Expected*)***

Dual Degree: B.Sc. in Mechanics & B.Eng. in Interdisciplinary Engineering

Core Courseworks:

Mathematical Analysis, Advanced linear Algebra, Probability and Statistics,
Data Structure and Algorithm Analysis, Machine Learning, Computer Organization Principle, Computer Networks

INTERNSHIP EXPERIENCES

Dongfeng Motor Corporation (R&D Institute) | AI Lab Intern ***Jun. 2025–Jul. 2025***

Project: Automated VQA Annotation for Real-Vehicle Data based on Impromptu-VLA & Senna

- Engineered an automated data preprocessing workflow using Python, converting vehicle logs into standard nuScenes format to ensure compatibility with open-source VLA frameworks.
- Designed a structural CoT prompting strategy that decomposes the annotation task into sequential logic.
- Implemented a "Single-View to Global Integration" reasoning mechanism with built-in hallucination checks.
- Optimized the inference architecture for resource-constrained environments by migrating the backbone from LLaVA-34B to a fine-tuned Senna-7B.
- Delivered a scalable, automated VQA annotation toolkit and a high-quality dataset containing rich multi-modal annotations.

RESEARCH EXPERIENCES

Human-like Driving Agent based on Large Language Models | Research Assistant ***Feb. 2025–Present***

Advisor: Prof. Jiangtao Gong | Institute for AI Industry Research, Tsinghua University

- Advancing an ICRA 2025 accepted framework from simulation to real-world deployment, aiming to bridge the sim-to-real gap for LLM-based autonomous agents.
- Established a rigorous evaluation taxonomy by conducting manual annotation and consistency checks on extensive literature, effectively performing domain-specific feature engineering to define critical semantic scene attributes.
- Operationalized these attributes as a structured perception schema for the VLM, guiding the model to extract literature-backed features as the primary basis for the LLM's intelligence assessment.

LLM-powered Framework for Driving Intelligence Evaluation | Research Assistant ***Apr. 2024–Oct. 2024***

Advisor: Prof. Jiangtao Gong | Institute for AI Industry Research, Tsinghua University

- Designed a hierarchical evaluation framework utilizing LLM with CoT reasoning to assess autonomous driving behavior across operational, tactical, and strategic levels.
- Implemented simulation experiments on CARLA Leaderboard 2.0; processed multi-modal driving data to validate the framework's effectiveness.
- Conducted human evaluation studies by designing questionnaires and analyzing feedback from participants, demonstrating a high alignment between the framework's output and human judgment.
- Co-authored a paper accepted by **ICRA 2025** (IEEE International Conference on Robotics and Automation)**.**

SELECTED PROJECT

Multi-Threaded FTP System with Resume Capability ***Nov. 2025***

- Developed a multi-threaded Linux FTP server to handle high-concurrency client connections efficiently.
- Implemented core FTP protocols including Active/Passive modes and breakpoint resume for reliable file transfers.
- Enforced strict security measures by restricting file paths to prevent directory traversal and unauthorized access.

HONORS & AWARDS

Science and Technology Innovation Excellence Scholarship(Top 10%), Xingjian College ***Nov. 2025***

Academic Excellence Scholarship(Top 5%), Tsinghua University ***Dec. 2024***

SKILLS

Programming Languages: C, C++, Python, Java, R
Others: LaTeX